Artemy Urodovskikh

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github.com/R2BPW · r2bpw.works

Education

MSc in Computer Science *2022*
Moscow Institute of Physics and Technology (MIPT).
Thesis: Visualization algorithms for large-scale route data on maps.

BSc in Software Engineering *2020*
MIREA — Russian Technological University.

Professional Experience

Backend Team Lead *May 2018 — present*
Bolshaya Troika, Moscow. 8+ years.

Core product: regional-scale waste-management platform (Python/Django, PostgreSQL, ClickHouse) for government clients in Russia and Kazakhstan. Multi-tenant deployments with per-region data residency. Led a team of five engineers within a 40-developer codebase. Authored specifications, delegated implementation to engineers and AI agents, reviewed outputs and integrated them into production.

- Built the route-planning system from scratch: background task queue, synchronous solver for 10,000+ waypoints per district, 500+ per vehicle. Standalone TSP solver with graceful degradation; in production since early 2022, dispatching roughly 1,200 routes per day.
- Built the billing system from scratch: tariff calculation, accruals, invoice generation for 1M+ individual payers. Prerequisite-validation pipeline with chunked processing and barrier-style result collection; runs abort on invalid data with an error report instead of partial commits. Memory-optimized batch printing.
- Designed and implemented the planning subsystem: from one plan per waste site to one plan per container with grouping at route-building stage, accounting for vehicle properties; enables per-container plan-vs-actual comparison.
- Designed two-leg hauls with independent documentation per vehicle and cascading reconciliation when a vehicle drops off the route.
- Zero-downtime S3 storage migration between providers via field-level storage routing and a background path-rewriting job.
- Fleet telemetry: data synchronization layer between ClickHouse and PostGIS. GPS filtering heuristics (track splitting, segment merging, outlier rejection) recovering actual movement geometry from drifting streams. Integrated S3 storage for onboard camera footage.
- Resolved a race condition in load-balancer port allocation for telemetry services; implemented auto-block for vehicles missing required attributes over a 7-day window.
- Hot operational tables exceed 500M rows; the telemetry pipeline ingests ~370M GPS samples per month from 5.7k active trackers into a 25-billion-sample ClickHouse archive.
- Non-blocking index creation on production tables with millions of rows. Eliminated N+1 queries by pushing joins into the database layer. Replaced a nested-loop plan on a 280M-row table with a hash anti-join, resolving statement timeouts.

Logistics Algorithms and Visualization *2020–2021*
Large-scale geospatial data: vehicle tracks, nested territories, graph overlays at regional scale.

Hardware–Software Complex for Municipal Inventory *2018–2019*
Automated yard territory inventory system. Still serving original clients.

Teaching

- Teaching assistant, MUCTR (2020); MIPT (2022).
- Code reviewer and mentor, Yandex.Practicum, backend development track (2023–2024).

Technical Skills

Languages	Python, SQL, C
Backend	Django, Django REST Framework, Celery, PostgreSQL, PostGIS, ClickHouse, Redis
Infrastructure	Docker, Kubernetes, Terraform, GitLab CI, S3 (Yandex/MinIO), Nginx, Grafana
Domain Expertise	GIS, route optimization (TSP/CVRP), telemetry, billing, DBSCAN clustering, object versioning
Other	FPGA (Xilinx Spartan-6), SDR, IrDA, Git, pytest, Agile

Certifications and Courses

- Cisco CCNA (2019).
- Practical Reinforcement Learning — HSE, Coursera (2021).
- Big Data Essentials: HDFS, MapReduce and Spark RDD — Yandex, Coursera (2020).

Selected Projects

Radio Bargain Bot r2bpw.works

Price-intelligence system for the secondary ham radio market. Aggregates listings from 98 sources across 50+ countries, scores each deal using Bayesian bootstrap benchmarks, publishes verified bargains to 12 regional Telegram channels.

Custom SDR Receiver on FPGA GitHub

SDR receiver firmware for Xilinx Spartan-6 (OSA103 Mini), written from scratch in Verilog: NCO, hardware-multiplier mixer, decimating CIC filter.

IR Bridge — Ericsson MC218 ↔ Flipper Zero ↔ LLM r2bpw.works

Bidirectional infrared bridge from a 1999 Ericsson MC218 palmtop to a large language model. Flipper Zero decodes IrDA and relays via ESP32 to an LLM API.

30m QRSS Beacons r2bpw.works

Catalogue of identified QRSS beacons on the 30-metre band, built on a distributed network of KiwiSDR receivers worldwide.